

A052 · Structure and Sequence Guided Drug Repurposing Framework for Antimalarial Target Discovery

Presenter: Fatemeh Ensafitakaldani — University of Massachusetts, Boston

Authors: Fatemeh Ensafitakaldani, Mohammad Hemmati, Kourosh Zarringham, Nurit Haspel

Session: Day 2 · Fri Oct 2, 2026 · 2:15–4:15 PM

Malaria remains one of the deadliest infectious diseases, causing over 200 million cases and roughly 600,000 deaths annually. Rising drug resistance in *Plasmodium falciparum* makes new therapeutics increasingly urgent, but discovery is hampered by low hit rates in high-throughput screens and limited understanding of which parasite proteins are druggable. We present a computational pipeline for drug repurposing that leverages existing drug-protein interaction data to prioritize candidate antimalarial compounds. Starting from active compounds identified through growth-inhibition screening against *P. falciparum*, we identify known protein targets and assess sequence and structural similarity between these targets and parasite proteins. Drugs whose targets show strong homology to a parasite protein are flagged as potential binders and further validated through structural modeling and binding-energy analysis. In parallel, we apply the pipeline in reverse-search mode across a druggability-informed map of the *P. falciparum* proteome, generating AlphaFold 3 structural models where experimental structures are unavailable. We test the pipeline's generalizability on an independent set of 50 compounds against 541 known druggable parasite proteins. This work produces a ranked, structure-informed map of the druggable *Plasmodium* proteome and experimentally supported compound-target relationships, including high-confidence drug-protein pairs. Future work will scale this approach to libraries of thousands of compounds, systematically mapping interactions and binding sites across the druggable malaria proteome. A *Plasmodium*-tuned structural tool as an HPC-ready repository will be provided.

Structure and Sequence Guided Drug Repurposing Framework for Antimalarial Target Discovery

Fatemeh Ensafitakaldani*, Mohammad Hemmati, Kourosh Zarringhalam, Nurit Haspel
University of Massachusetts, Boston

Rising drug resistance in *Plasmodium falciparum* makes new antimalarial therapeutics increasingly urgent, yet discovery is hampered by low screening hit rates and limited knowledge of druggable parasite proteins. We present a computational drug-repurposing pipeline that leverages existing drug-protein interaction data to prioritize candidate antimalarial compounds. Starting from active compounds identified through growth-inhibition screening against *P. falciparum*, we identify their known protein targets and assess sequence and structural similarity between these targets and *P. falciparum* proteins. Drugs whose targets show strong homology to parasite proteins are prioritized as potential binders, with predicted interactions further evaluated through structural modeling and binding-energy analysis. This approach yields high-confidence drug–protein pairs linking known drugs to specific, previously unmapped malaria targets.

In parallel, we apply the pipeline in reverse-search mode to a curated set of ~50 drugs and a druggability-informed map of the *P. falciparum* proteome comprising 541 known druggable proteins from prior structure-based druggability mining. This identifies the parasite proteins each drug is most likely to engage. Given the limited availability of experimentally resolved *Plasmodium* structures, we generate models for relevant proteins using AlphaFold 3, enabling structural comparison and interaction analysis for previously uncharacterized targets. This provides additional validation of the pipeline and expands high-confidence drug–protein predictions across a broader, structure-informed proteome map.

Building on this framework, we plan to scale the approach to thousands of compounds, systematically mapping predicted interactions and binding sites across the druggable *Plasmodium* proteome. To date, the work has delivered a ranked, structure and function-informed map of the druggable *Plasmodium* proteome. We have also generated an experimentally supported compound-target set of relationships, with high-confidence drug-protein pairs from initial validation. A *Plasmodium*-tuned structural tool as an HPC-ready repository will be provided as our future work.

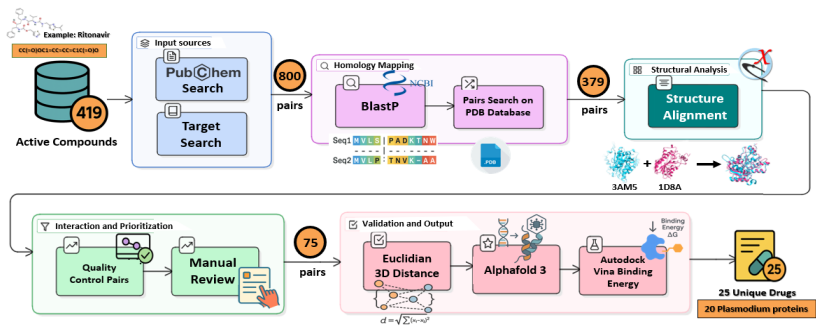


Figure 1. Computational Drug-repurposing Pipeline. Identifying known protein targets for the initial drug dataset and assessing sequence and structural similarity between these targets and *P. falciparum* proteins.